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MARI — Detailed Weekly Report

Week 3: Real Competitor Baselines (ART, PGM, Bx-tree)

Member A (Theory & Writing) • Member B (Implementation & Evaluation)

1. Objective

Compare MARI head-to-head against faithful, oracle-verified ordered indexes on the same exact-range-under-drift problem.

2. Motivation — why this week

The strongest reviewer objection is the absence of a competitive baseline. We answer it with real indexes: an adaptive radix tree, a dynamic learned index, and a Bx-tree made exact by a verification adapter.

3. Method & key equations

Relocations/update: MARI 0.024 vs 0.990 $\Rightarrow \sim 40 \times$ ($\sim 110 \times$ at scale).

Exactness tax: ART full relocation; PGM $\sim 5 \times$ merge writes; Bx $\sim 7 \times$ verify/result.

4. Process (step by step)

1. Implement ART (Node4/16/48/256), a dynamic PGM (sorted level-0 buffer + verification + merge-GC), and a Bx-tree + exactness adapter in `competitors.py`.
2. Verify each competitor exact against the oracle.
3. Run `python baseline_bench.py`: 5 seeds plus a scale run, recording relocation/write/verify counts.
4. Plot the comparison with `python fig_baseline.py`.

5. Results

Structure	Reloc / upd	Writes / upd	Verify / result	Exact?
MARI	0.024	1.15	1.01	yes
ART	0.990	1.08	0.00	yes
PGM	0.990	4.96	1.38	yes
Bx-tree	0.990	1.16	6.93	yes

- MARI relocates $\sim 40 \times$ less than every competitor ($\sim 110 \times$ at scale); all four exact.
- Each competitor stays exact only by paying elsewhere (ART relocation, PGM merge writes, Bx verification).

6. Interpretation

The comparison locates MARI's value rather than asserting it: the relocation rate is the definitional contrast, and each competitor's 'exactness tax' shows there is no free lunch without the guard.

7. Who did what

- **Member A:** Defined the shared problem and fairness framing (implementation-independent metrics; the Bx adapter is our construction); wrote the comparison section.
- **Member B:** Implemented all three competitors, verified exactness, ran the 5-seed + scale benchmark, drew the figure.

8. Deliverables (code)

`competitors.py`, `baseline_bench.py`, `fig-baseline.py`

9. Status

Status: Completed.

10. Next week

Week 4 checks whether bounded drift is real across independent domains. _____

Future work, beyond this plan: a native, GIL-free multicore implementation benchmarked against tuned native indexes (ART, Bw-tree, ALEX/PGM, Bx-tree) on real hardware — the gate to a top-tier venue. It needs hardware, not more design.